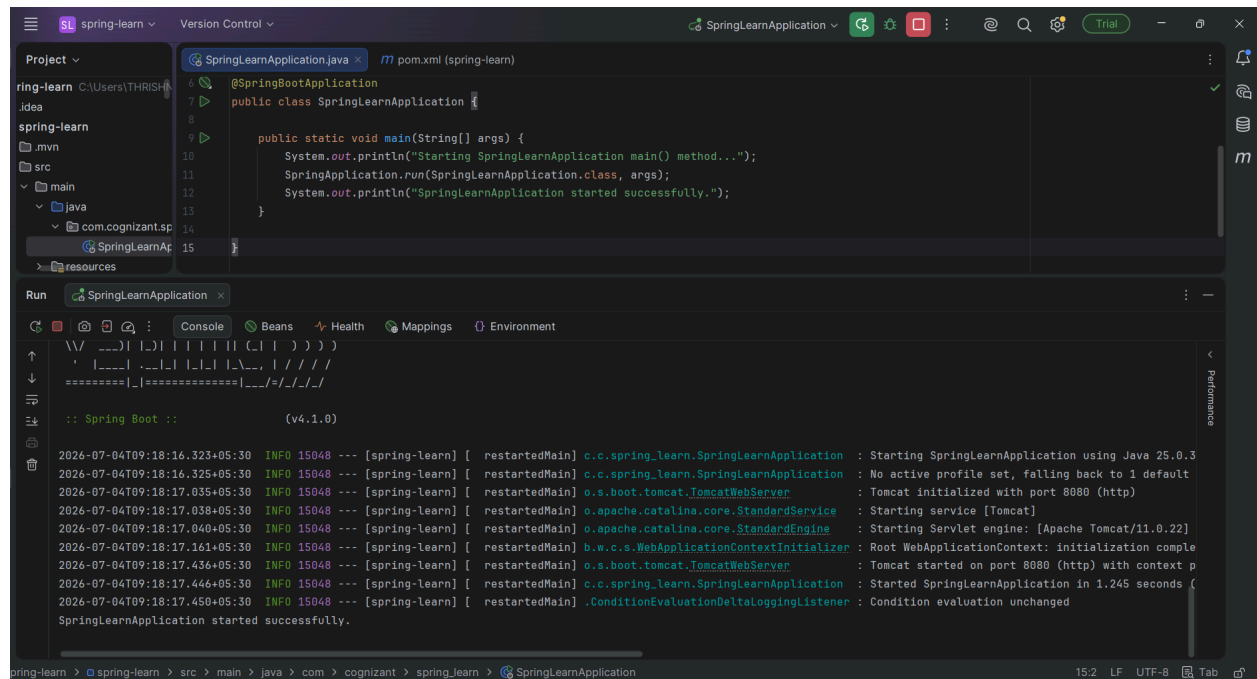


Spring REST using Spring Boot 3

1. Spring-rest-handson

Create a Spring Web Project using Maven



The screenshot displays an IDE window for a project named 'spring-learn'. The main editor shows the `SpringLearnApplication.java` file, which is a `@SpringBootApplication` class. The `main` method prints messages indicating the start and successful completion of the application.

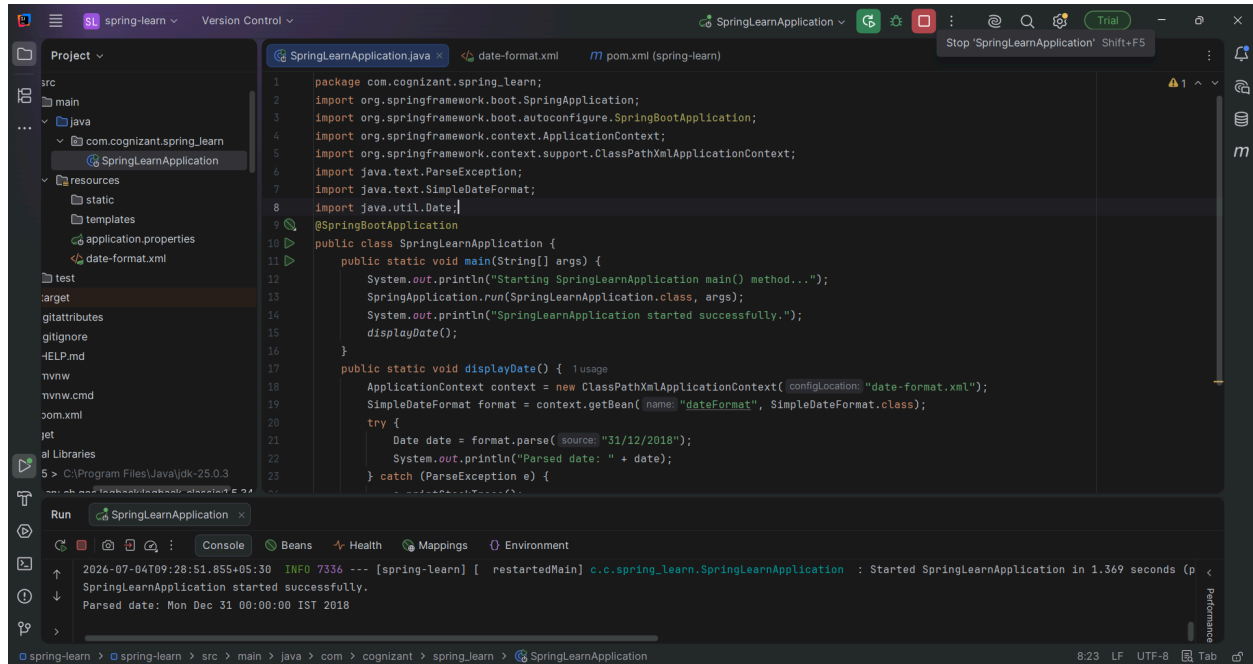
```
1 @SpringBootApplication
2 public class SpringLearnApplication {
3
4     public static void main(String[] args) {
5         System.out.println("Starting SpringLearnApplication main() method...");
6         SpringApplication.run(SpringLearnApplication.class, args);
7         System.out.println("SpringLearnApplication started successfully.");
8     }
9 }
```

The Run console shows the output of the application, including the Spring Boot logo and various log messages from the `restarterMain` thread. The logs indicate that the application is starting using Java 25.0.3, no active profile is set, and the application starts successfully on port 8080.

```
2026-07-04T09:18:16.323+05:30 INFO 15048 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.3
2026-07-04T09:18:16.325+05:30 INFO 15048 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default
2026-07-04T09:18:17.038+05:30 INFO 15048 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-04T09:18:17.040+05:30 INFO 15048 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-04T09:18:17.040+05:30 INFO 15048 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T09:18:17.161+05:30 INFO 15048 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complete
2026-07-04T09:18:17.436+05:30 INFO 15048 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path [/]
2026-07-04T09:18:17.446+05:30 INFO 15048 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.245 seconds
2026-07-04T09:18:17.450+05:30 INFO 15048 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition evaluation unchanged
SpringLearnApplication started successfully.
```

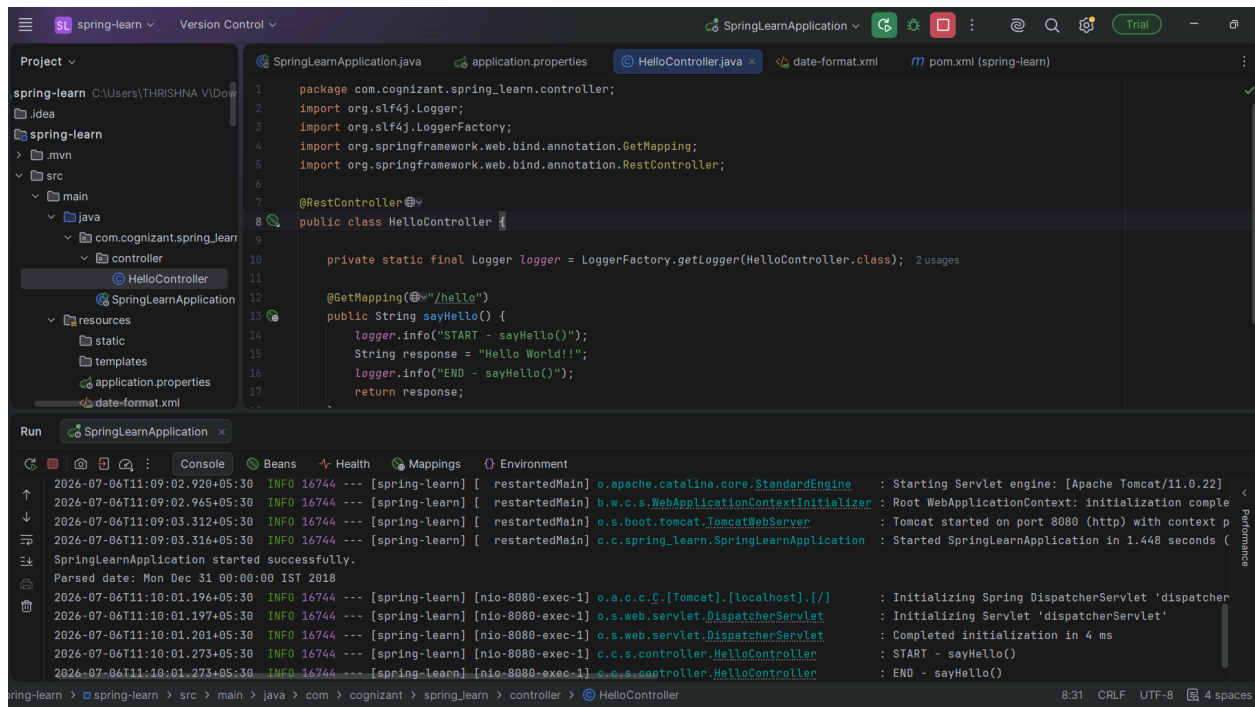
1. Spring-rest-handson

Spring Core – Load SimpleDateFormat from Spring Configuration XML



2. spring-rest-handson

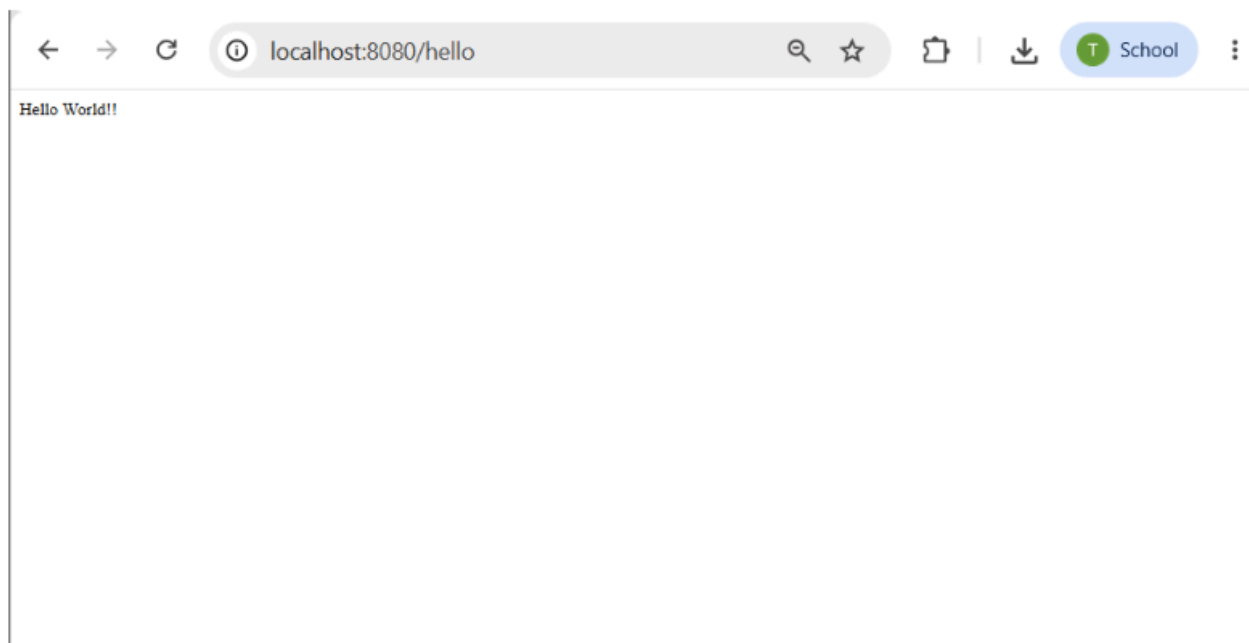
Hello World RESTful Web Service



The screenshot shows an IDE with the following components:

- Project Explorer:** Shows the project structure for 'spring-learn' with folders like 'main', 'resources', and 'static'.
- Code Editor:** Displays the 'HelloController.java' file with the following code:

```
1 package com.cognizant.spring_learn.controller;
2 import org.slf4j.Logger;
3 import org.slf4j.LoggerFactory;
4 import org.springframework.web.bind.annotation.GetMapping;
5 import org.springframework.web.bind.annotation.RestController;
6
7 @RestController
8 public class HelloController {
9
10     private static final Logger logger = LoggerFactory.getLogger(HelloController.class);
11
12     @GetMapping("/hello")
13     public String sayHello() {
14         logger.info("START - sayHello()");
15         String response = "Hello World!!";
16         logger.info("END - sayHello()");
17         return response;
18     }
19 }
```
- Run Console:** Shows the application's startup logs, including the message 'SpringLearnApplication started successfully' and the response 'Hello World!!'.



GET http://localhost:8083/hello

HTTP

http://localhost:8083/hello

Save

GET

http://localhost:8083/hello

Send

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Cookies

Query Params

	Key	Value	Bulk Edit
	Key	Value	

Body

Cookies

Headers (5)

Test Results

Status: 200 OK Time: 441 ms Size: 177 B Save Response

Pretty

Raw

Preview

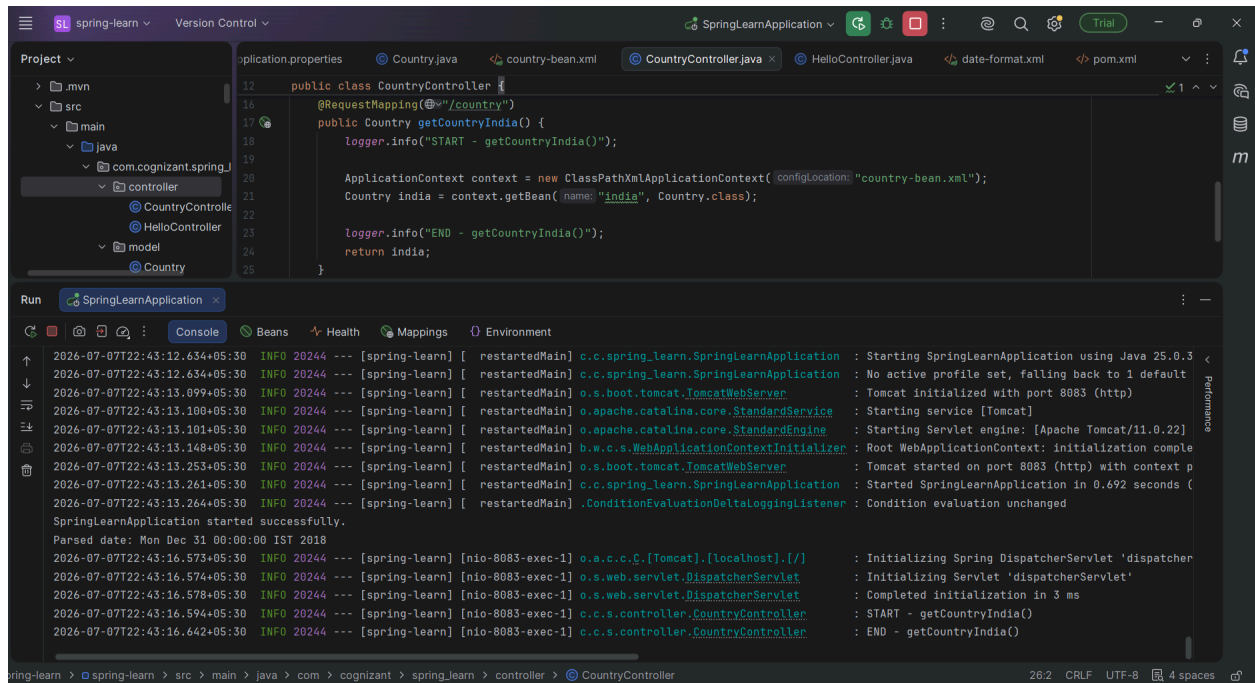
Visualize

Text

1 Hello World!!

2. spring-rest-handson

REST - Country Web Service



The screenshot shows an IDE with the following components:

- Project Explorer:** Shows the project structure with folders for `src`, `main`, `java`, and `com.cognizant.spring.learn`. The `controller` folder contains `CountryController`, `HelloController`, and `Country`.
- Editor:** Displays the `CountryController.java` file. The code is as follows:

```
12 public class CountryController {  
16     @RequestMapping("/country")  
17     public Country getCountryIndia() {  
18         logger.info("START - getCountryIndia()");  
19  
20         ApplicationContext context = new ClassPathXmlApplicationContext("country-bean.xml");  
21         Country india = context.getBean("india", Country.class);  
22  
23         logger.info("END - getCountryIndia()");  
24         return india;  
25     }  
}
```
- Run Console:** Shows the output of the application. The logs indicate that the application started successfully, initialized the Tomcat server, and the `getCountryIndia()` method was called, returning the country information.

```
2026-07-07T22:43:12.634+05:30 INFO 20244 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 25.0.3  
2026-07-07T22:43:12.634+05:30 INFO 20244 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default  
2026-07-07T22:43:13.099+05:30 INFO 20244 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)  
2026-07-07T22:43:13.104+05:30 INFO 20244 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]  
2026-07-07T22:43:13.104+05:30 INFO 20244 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]  
2026-07-07T22:43:13.148+05:30 INFO 20244 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complete  
2026-07-07T22:43:13.253+05:30 INFO 20244 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path  
2026-07-07T22:43:13.261+05:30 INFO 20244 --- [spring-learn] [ restartedMain] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.692 seconds (process running for 1.011 seconds)  
2026-07-07T22:43:13.264+05:30 INFO 20244 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition evaluation unchanged  
SpringLearnApplication started successfully.  
Parsed date: Mon Dec 31 00:00:00 IST 2018  
2026-07-07T22:43:16.573+05:30 INFO 20244 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'  
2026-07-07T22:43:16.574+05:30 INFO 20244 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'  
2026-07-07T22:43:16.578+05:30 INFO 20244 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms  
2026-07-07T22:43:16.594+05:30 INFO 20244 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.CountryController : START - getCountryIndia()  
2026-07-07T22:43:16.642+05:30 INFO 20244 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.CountryController : END - getCountryIndia()
```

localhost:8083/country

Pretty-print ☐

```
{"code": "IN", "name": "India"}
```

GET http://localhost:8083/cou

+

...

http

http://localhost:8083/country

Save

GET

⌵

http://localhost:8083/country

Send

⌵

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Cookies

Query Params

	Key	Value	Bulk Edit
	Key	Value	

Body

Cookies

Headers (5)

Test Results

⌕

Status: 200 OK

Time: 53 ms

Size: 184 B

Save Response

⌵

Pretty

Raw

Preview

Visualize

JSON

⌵

⌵

⌵

⌵

⌵

1

2

3

4

⌵

⌵

⌵

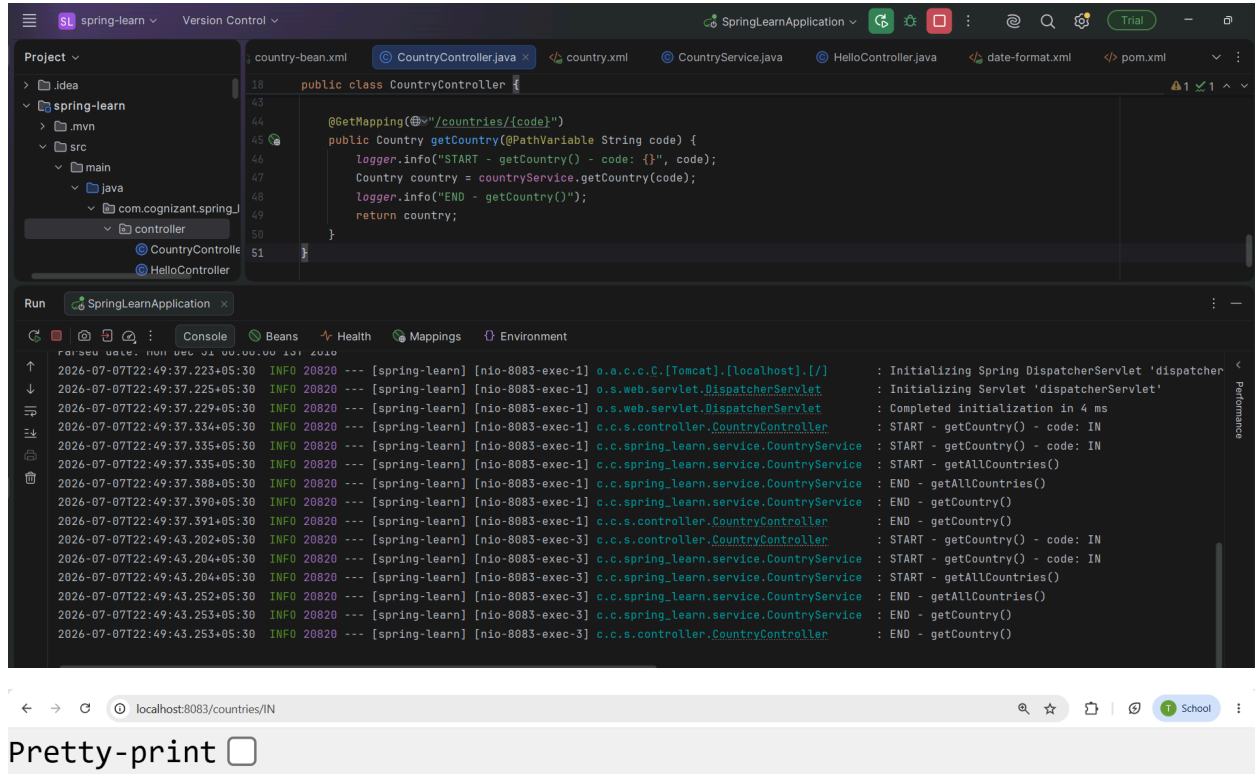
⌵

"code": "IN",

"name": "India"

2. Spring-rest-handson

REST - Get country based on country code



The screenshot displays a development environment with an IDE and a web browser. The IDE shows the `CountryController.java` file with the following code:

```
18 public class CountryController {
43
44     @GetMapping("/countries/{code}")
45     public Country getCountry(@PathVariable String code) {
46         logger.info("START - getCountry() - code: {}", code);
47         Country country = countryService.getCountry(code);
48         logger.info("END - getCountry()");
49         return country;
50     }
51 }
```

The Run console shows the following logs:

```
2026-07-07T22:49:37.223+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcher
2026-07-07T22:49:37.225+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-07T22:49:37.229+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 4 ms
2026-07-07T22:49:37.334+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.CountryController : START - getCountry() - code: IN
2026-07-07T22:49:37.335+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] c.c.spring_learn.service.CountryService : START - getCountry() - code: IN
2026-07-07T22:49:37.388+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] c.c.spring_learn.service.CountryService : END - getAllCountries()
2026-07-07T22:49:37.390+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] c.c.spring_learn.service.CountryService : END - getCountry()
2026-07-07T22:49:37.391+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-1] c.c.s.controller.CountryController : END - getCountry()
2026-07-07T22:49:43.202+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.CountryController : START - getCountry() - code: IN
2026-07-07T22:49:43.204+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.spring_learn.service.CountryService : START - getCountry() - code: IN
2026-07-07T22:49:43.252+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.spring_learn.service.CountryService : START - getAllCountries()
2026-07-07T22:49:43.253+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.spring_learn.service.CountryService : END - getAllCountries()
2026-07-07T22:49:43.253+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.spring_learn.service.CountryService : END - getCountry()
2026-07-07T22:49:43.253+05:30 INFO 20820 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.CountryController : END - getCountry()
```

The browser shows the URL `localhost:8083/countries/IN` and the response:

```
Pretty-print {"code": "IN", "name": "India"}
```

GET http://localhost:8083/cou

+

...

HTTP

http://localhost:8083/countries/IN

Save

</>

GET

▼

http://localhost:8083/countries/IN

Send

▼

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Cookies

Query Params

	Key	Value	Bulk Edit
	Key	Value	

Body

Cookies

Headers (5)

Test Results

🌐

Status: 200 OK

Time: 69 ms

Size: 184 B

Save Response

▼

Pretty

Raw

Preview

Visualize

JSON

▼

1

2

3

4

1

2

3

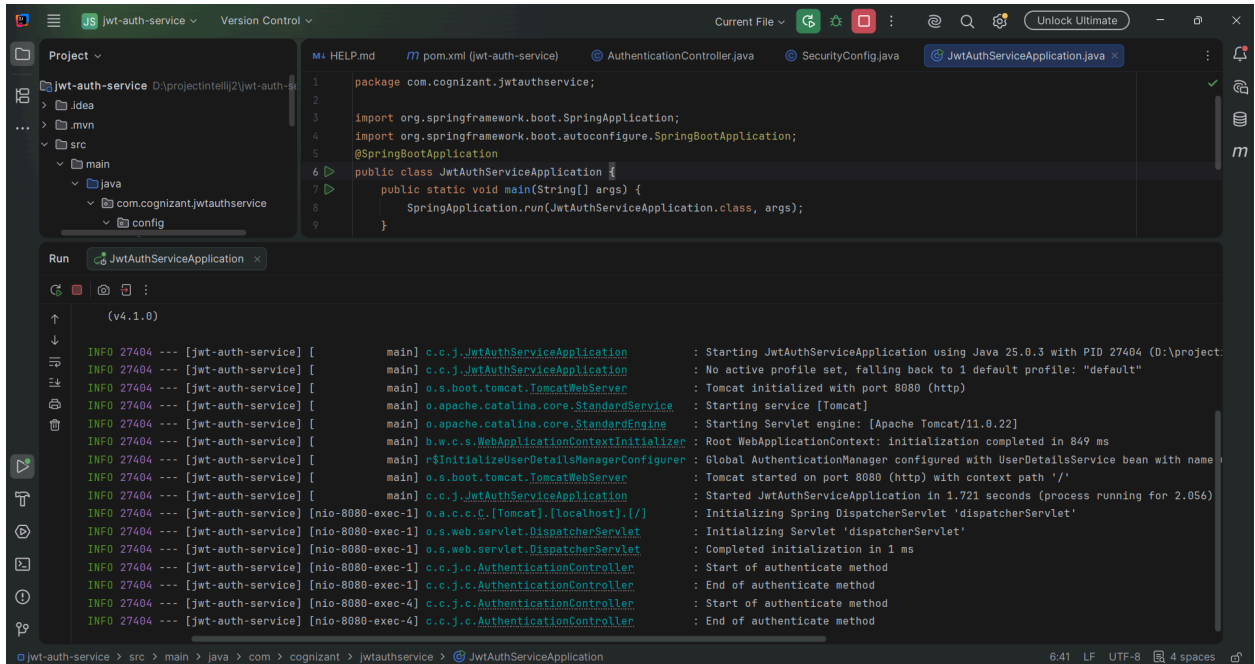
4

"code": "IN",

"name": "India"

5. JWT-handson

Create authentication service that returns JWT



```
package com.cognizant.jwtauthservice;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

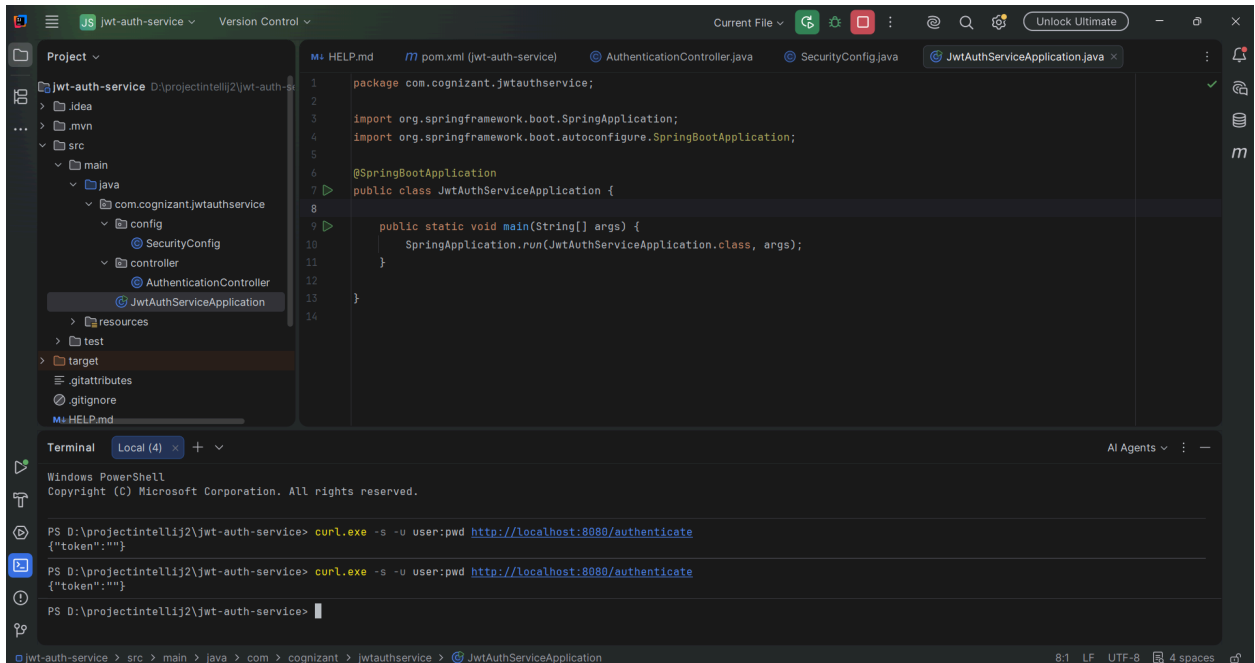
@SpringBootApplication
public class JwtauthServiceApplication {

    public static void main(String[] args) {
        SpringApplication.run(JwtauthServiceApplication.class, args);
    }
}
```

Run JwtauthServiceApplication

(v4.1.0)

```
INFO 27404 --- [jwt-auth-service] [main] c.c.j.JwtauthServiceApplication : Starting JwtauthServiceApplication using Java 25.0.3 with PID 27404 (0:\project\
INFO 27404 --- [jwt-auth-service] [main] c.c.j.JwtauthServiceApplication : No active profile set, falling back to 1 default profile: "default"
INFO 27404 --- [jwt-auth-service] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
INFO 27404 --- [jwt-auth-service] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
INFO 27404 --- [jwt-auth-service] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
INFO 27404 --- [jwt-auth-service] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 849 ms
INFO 27404 --- [jwt-auth-service] [main] r$InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsService bean with name
INFO 27404 --- [jwt-auth-service] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
INFO 27404 --- [jwt-auth-service] [main] c.c.j.JwtauthServiceApplication : Started JwtauthServiceApplication in 1.721 seconds (process running for 2.056)
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-1] c.c.j.c.AuthenticationController : Start of authenticate method
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-1] c.c.j.c.AuthenticationController : End of authenticate method
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-4] c.c.j.c.AuthenticationController : Start of authenticate method
INFO 27404 --- [jwt-auth-service] [nio-8080-exec-4] c.c.j.c.AuthenticationController : End of authenticate method
```



```
package com.cognizant.jwtauthservice;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class JwtauthServiceApplication {

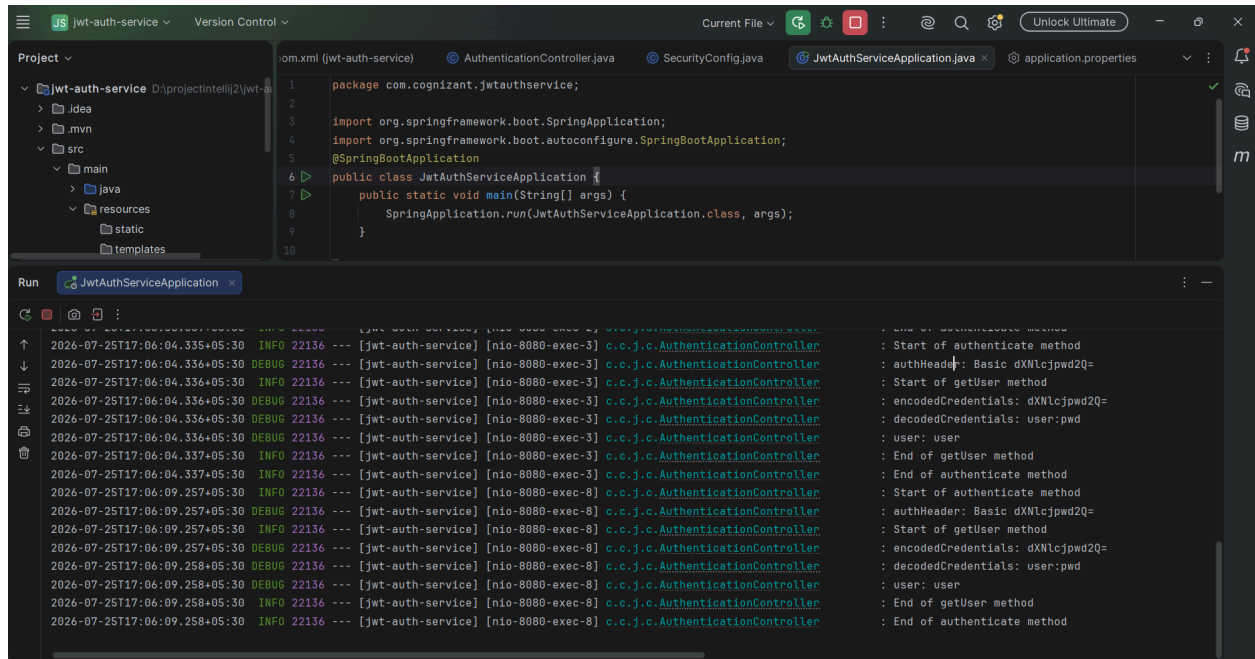
    public static void main(String[] args) {
        SpringApplication.run(JwtauthServiceApplication.class, args);
    }
}
```

Terminal Local (4)

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

```
PS D:\project\intelli2\jwt-auth-service> curl.exe -s -u user:pwd http://localhost:8080/authenticate
{"token":""}
PS D:\project\intelli2\jwt-auth-service> curl.exe -s -u user:pwd http://localhost:8080/authenticate
{"token":""}
PS D:\project\intelli2\jwt-auth-service>
```

Read Authorization header and decode the username and password



Generate token based on the user

